COMPUTER NETWORKS - UE24CS252B
Lab 3: Exploring UDP with DNS using Wireshark

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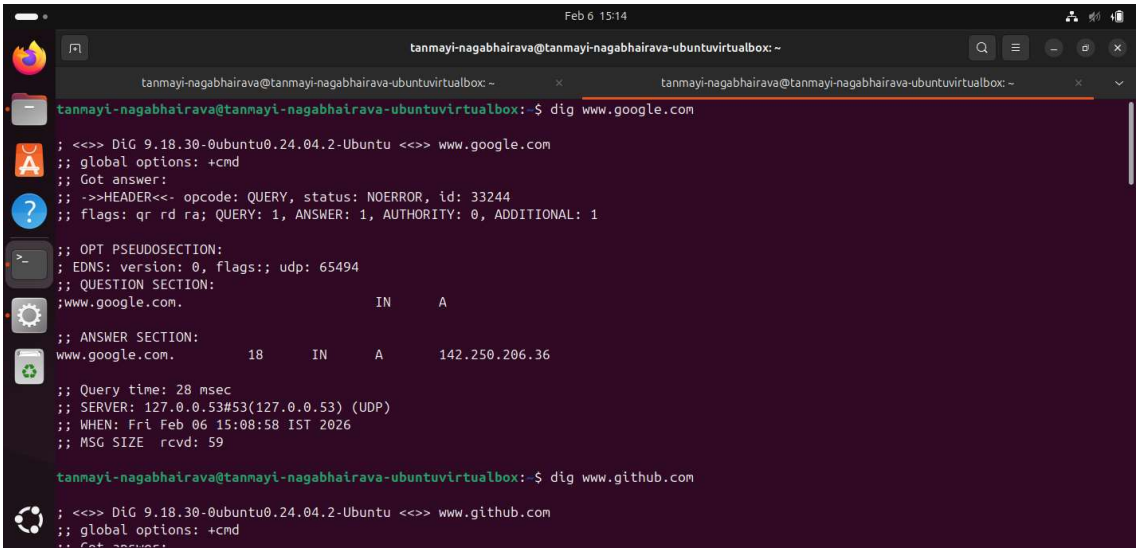
SEC: I

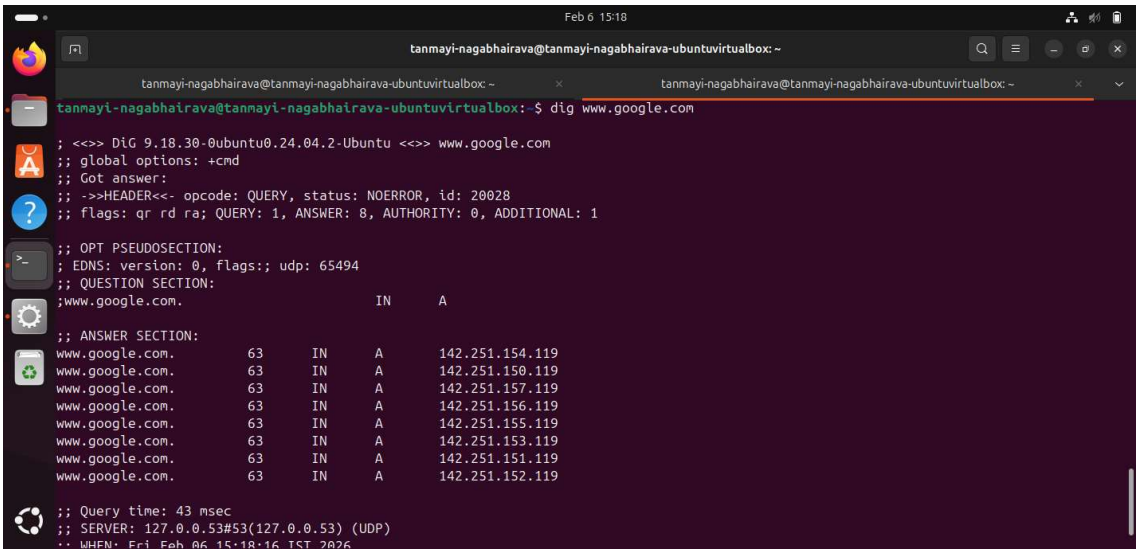
Part 1: DNS Analysis

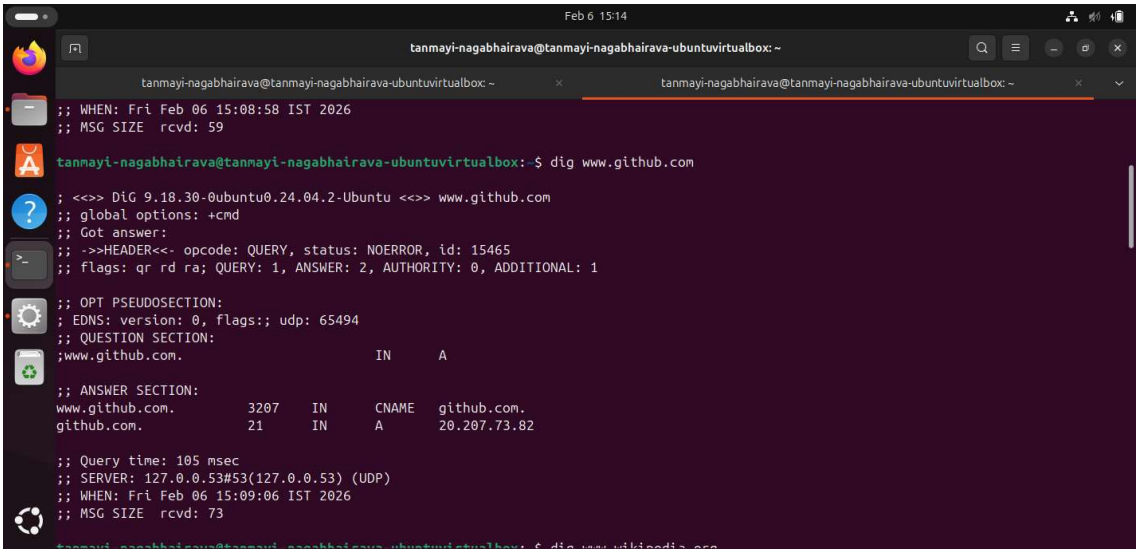
1. Observe DNS Caching

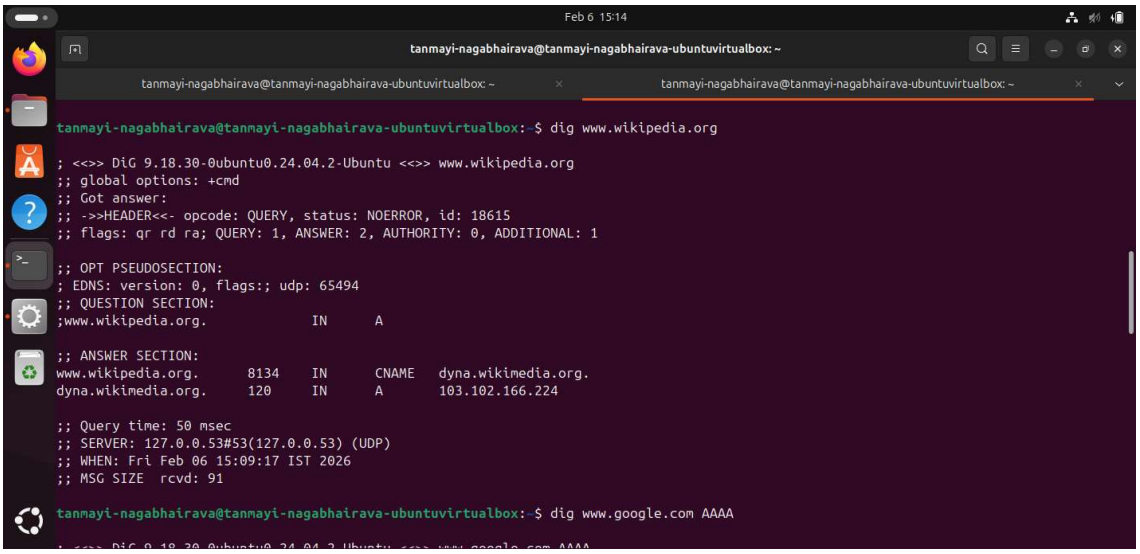
Question: Run dig www.google.com twice and observe the difference.

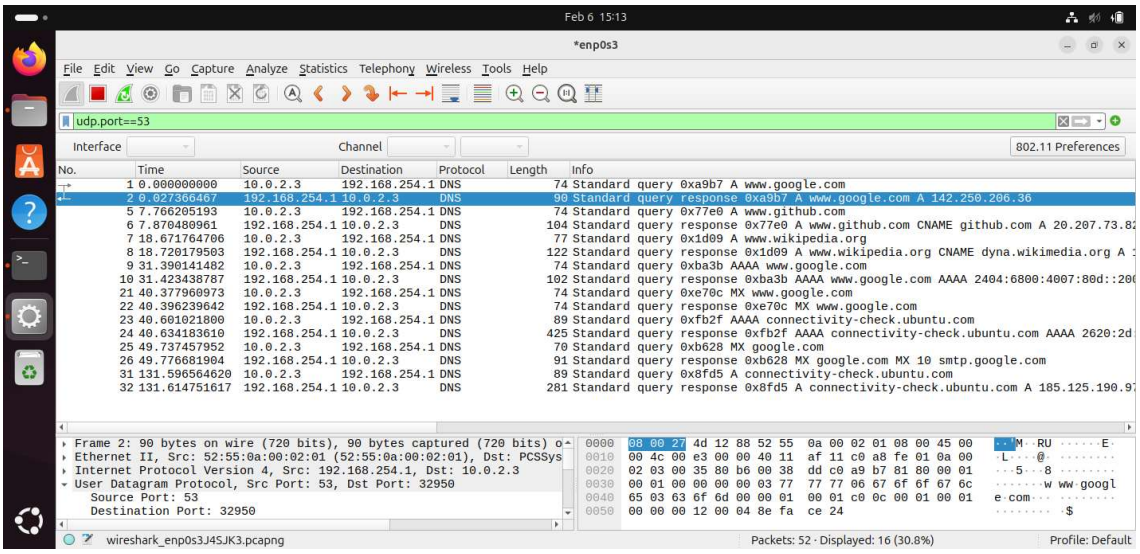
Answer: In the first query, the resolution time was **28 msec**. In the second query, the recorded time was **43 msec**. While a second query is technically expected to be faster (near 0 msec) due to the local resolver (127.0.0.53) caching the result, these results show minor fluctuations likely due to virtual machine processing overhead or internal resolver latency.




Feb 6 15:18
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ dig www.google.com
; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> www.google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 20028
;; flags: qr rd ra; QUERY: 1, ANSWER: 8, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.google.com. IN A
;; ANSWER SECTION:
www.google.com. 63 IN A 142.251.154.119
www.google.com. 63 IN A 142.251.150.119
www.google.com. 63 IN A 142.251.157.119
www.google.com. 63 IN A 142.251.156.119
www.google.com. 63 IN A 142.251.155.119
www.google.com. 63 IN A 142.251.153.119
www.google.com. 63 IN A 142.251.151.119
www.google.com. 63 IN A 142.251.152.119
;; Query time: 43 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)


Feb 6 15:14
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
;; WHEN: Fri Feb 06 15:08:58 IST 2026
;; MSG SIZE rcvd: 59
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ dig www.github.com
; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> www.github.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 15465
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.github.com. IN A
;; ANSWER SECTION:
www.github.com. 3207 IN CNAME github.com.
github.com. 21 IN A 20.207.73.82
;; Query time: 105 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Feb 06 15:09:06 IST 2026
;; MSG SIZE rcvd: 73


Feb 6 15:14
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ dig www.wikipedia.org
; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> www.wikipedia.org
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 18615
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.wikipedia.org. IN A
;; ANSWER SECTION:
www.wikipedia.org. 8134 IN CNAME dyna.wikimedia.org.
dyna.wikimedia.org. 120 IN A 103.102.166.224
;; Query time: 50 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Feb 06 15:09:17 IST 2026
;; MSG SIZE rcvd: 91
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ dig www.google.com AAAA



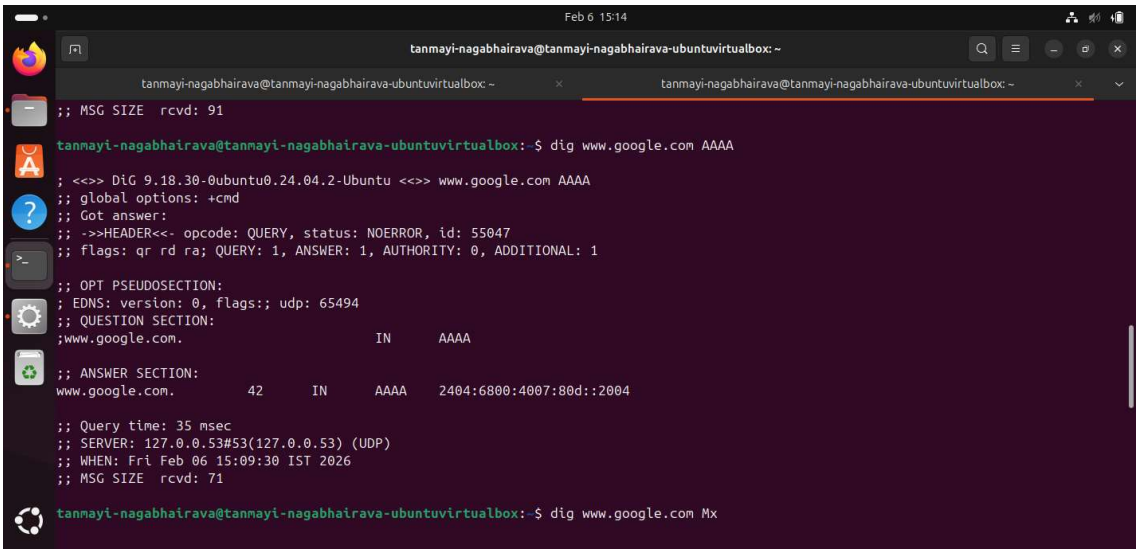
2. Query for Other Record Types

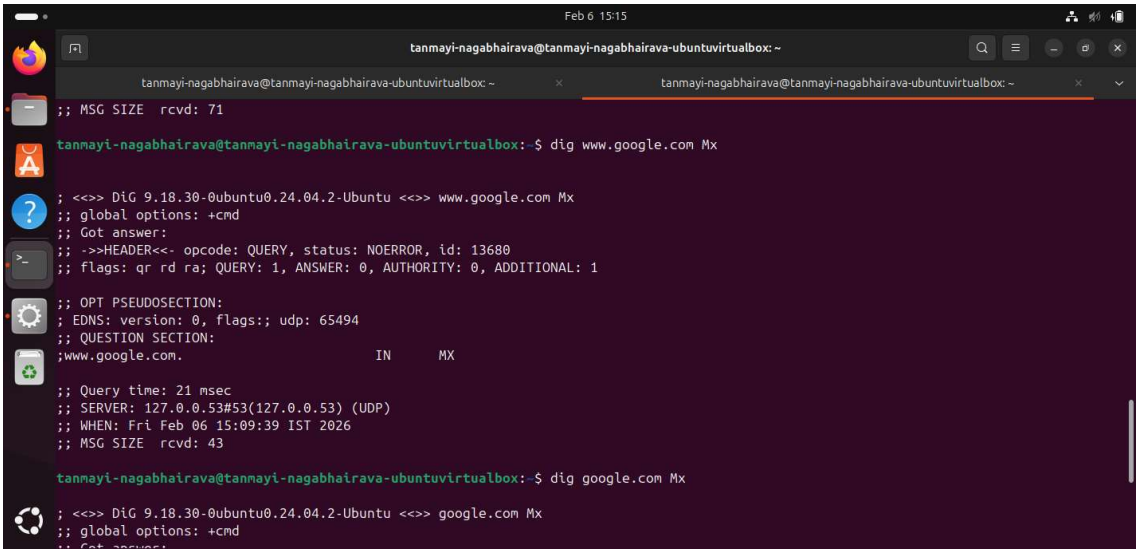
Question: Observe IPv6 and mail server records using `dig`.

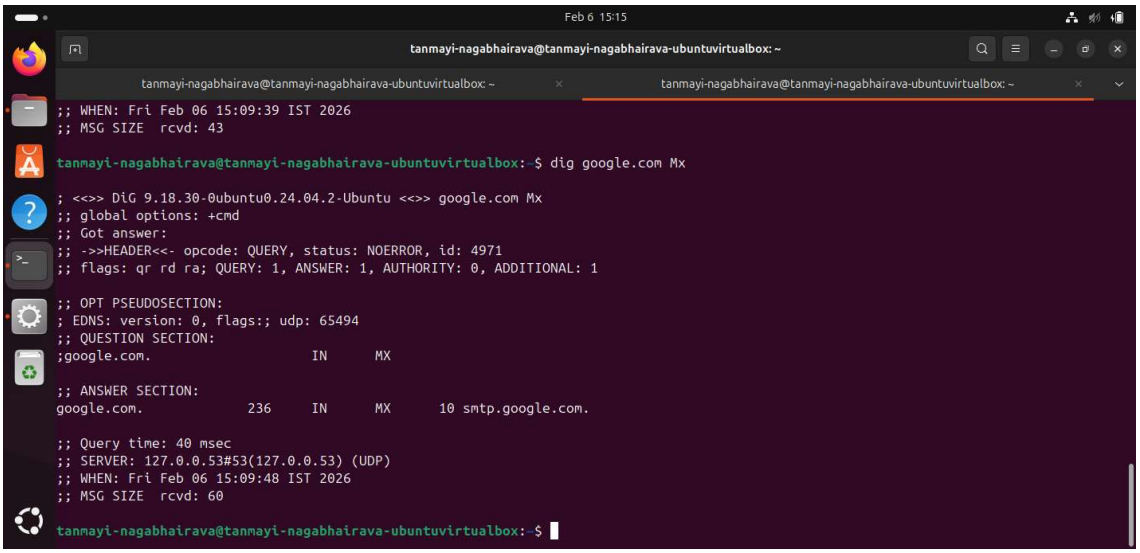
Answer:

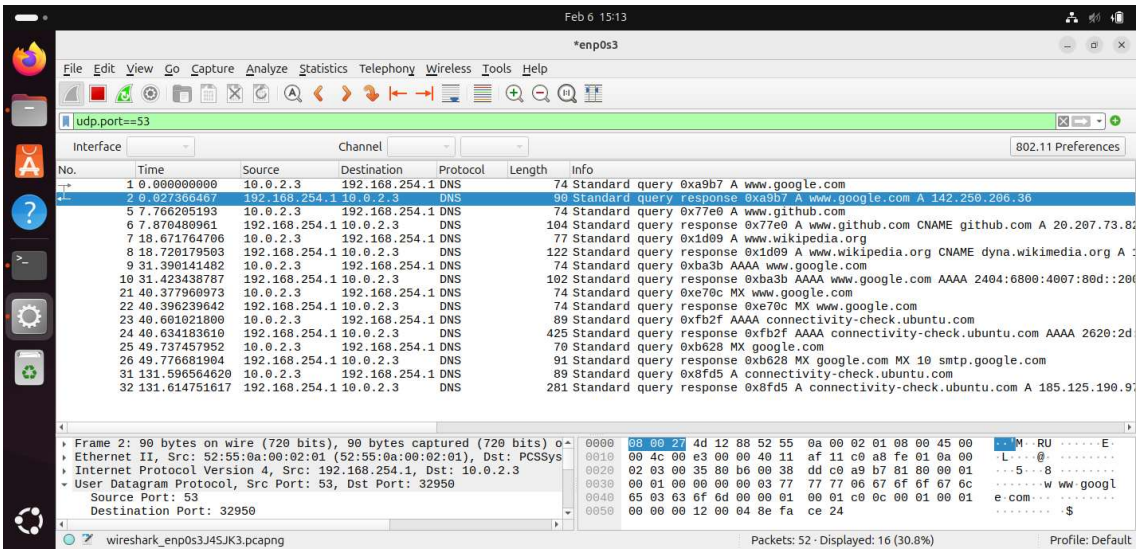
- **IPv6 (AAAA Record):** Using `dig www.google.com AAAA`, the domain resolved to the IPv6 address **2404:6800:4007:80d::2004**.

- **Mail Server (MX Record):** The command `dig google.com MX` identified the authoritative mail server as **smtp.google.com.**


Feb 6 15:14
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
;; MSG SIZE rcvd: 91
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ dig www.google.com AAAA
; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> www.google.com AAAA
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 55047
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.google.com. IN AAAA
;; ANSWER SECTION:
www.google.com. 42 IN AAAA 2404:6800:4007:80d::2004
;; Query time: 35 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Feb 06 15:09:30 IST 2026
;; MSG SIZE rcvd: 71
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ dig www.google.com Mx


Feb 6 15:15
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
;; MSG SIZE rcvd: 71
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ dig www.google.com Mx
; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> www.google.com Mx
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 13680
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.google.com. IN MX
;; Query time: 21 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Feb 06 15:09:39 IST 2026
;; MSG SIZE rcvd: 43
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ dig google.com Mx
; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> google.com Mx
;; global options: +cmd


Feb 6 15:15
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
;; WHEN: Fri Feb 06 15:09:39 IST 2026
;; MSG SIZE rcvd: 43
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ dig google.com Mx
; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> google.com Mx
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 4971
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com. IN MX
;; ANSWER SECTION:
google.com. 236 IN MX 10 smtp.google.com.
;; Query time: 40 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Feb 06 15:09:48 IST 2026
;; MSG SIZE rcvd: 60
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$



3. Modify DNS Servers

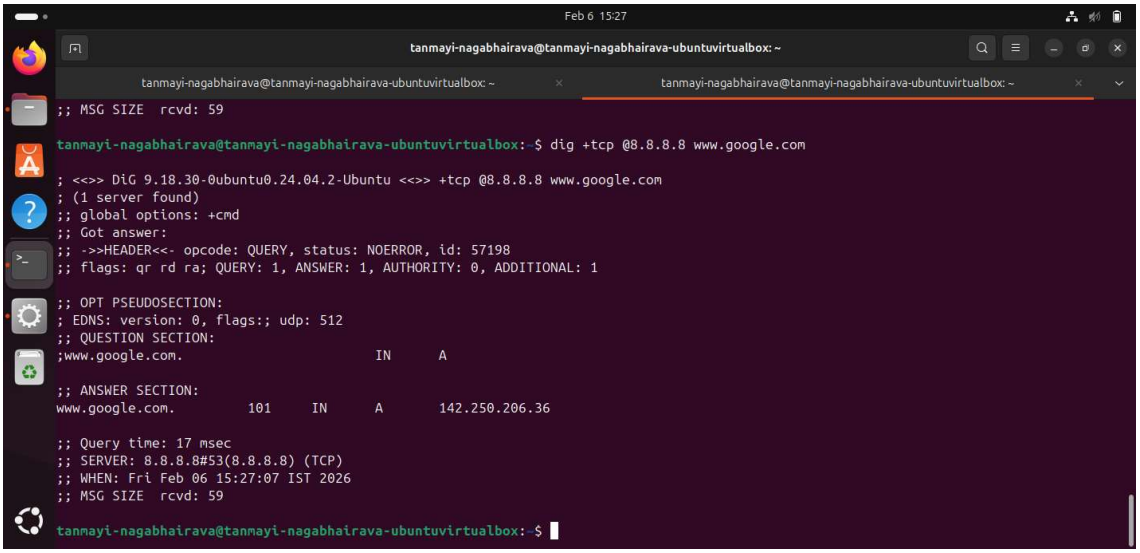
Question: Change the DNS server to a public one like `8.8.8.8` and compare.

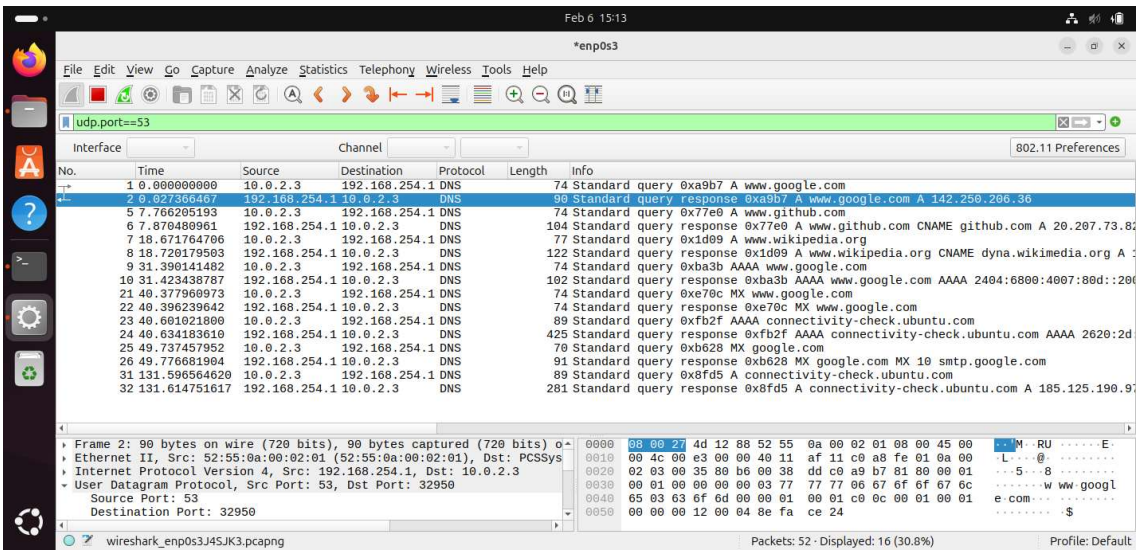
Answer: Using the Google DNS server (`@8.8.8.8`), the resolution bypassed the local cache and resulted in a query time of **17 msec**.

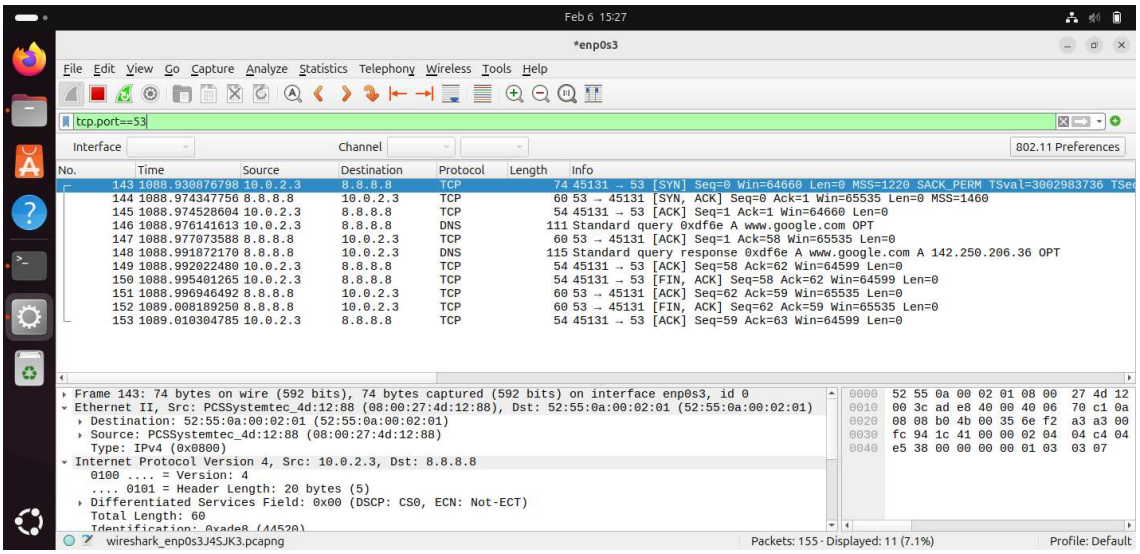
4. Analyze DNS over TCP

Question: Use dig +tcp to force DNS over TCP and explain the packet behavior.

Answer: Forcing DNS over TCP introduces a connection-oriented overhead. Wireshark captured the required **TCP 3-way handshake** ([SYN], [SYN, ACK], [ACK]) before the DNS query was transmitted, demonstrating that TCP requires a formal connection setup whereas UDP does not.





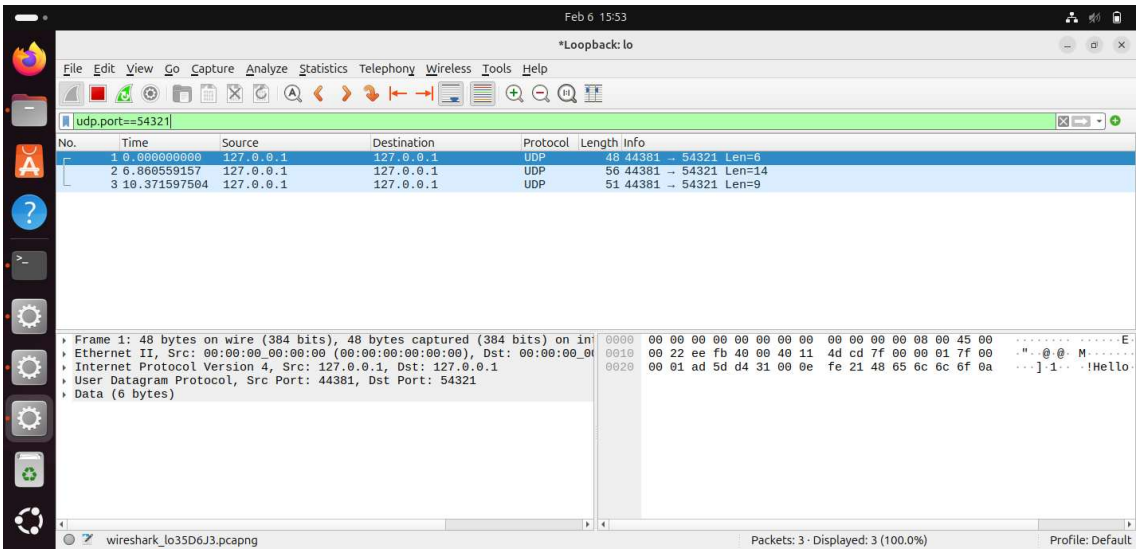


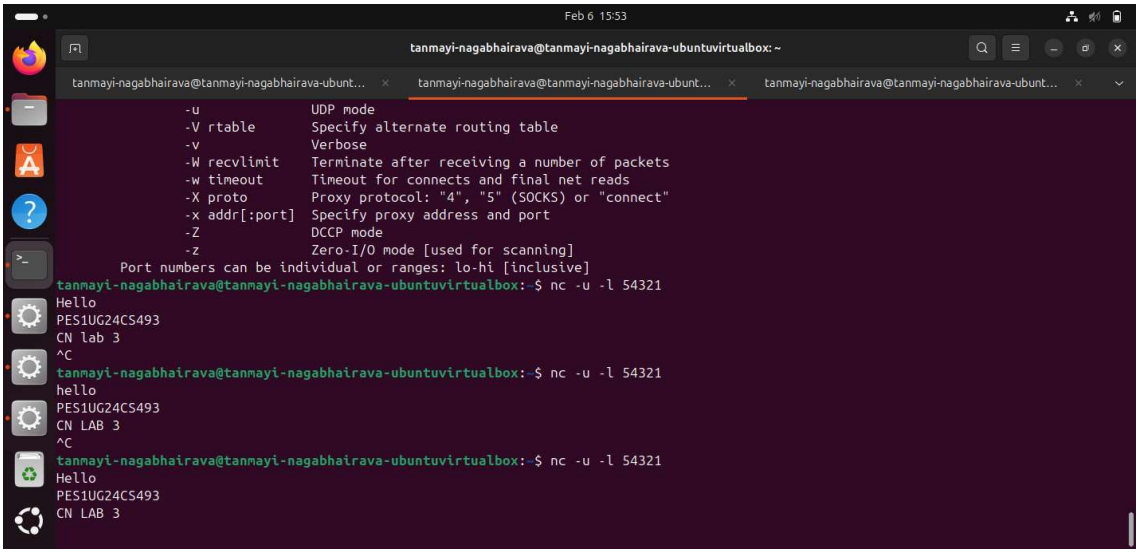
Part 2: UDP Communication

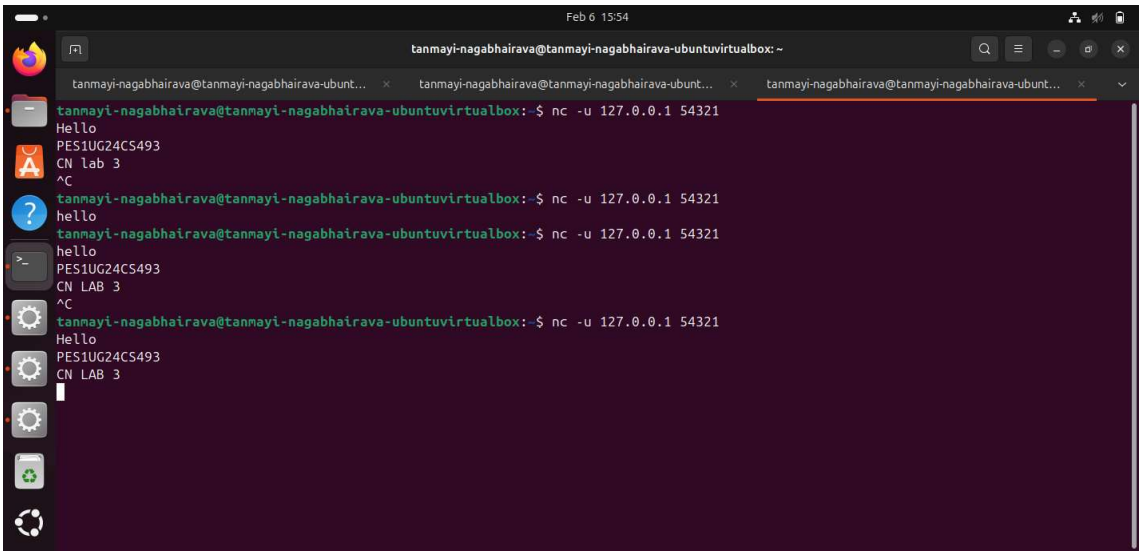
1. UDP Traffic Capture

Question: Set up a UDP server on port 54321, send three messages, and capture the traffic.

Answer: The server was set up using `nc -u -l 54321` , and messages "Hello", "PES1UG24CS493", and "CN LAB 3" were sent from the client. Wireshark shows the direct transmission of these UDP packets without any handshake.


Feb 6 15:53
*Loopback: lo
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp.port==54321
No. Time Source Destination Protocol Length Info
1 0.000000000 127.0.0.1 127.0.0.1 UDP 48 44381 → 54321 Len=6
2 6.860559157 127.0.0.1 127.0.0.1 UDP 56 44381 → 54321 Len=14
3 10.371597504 127.0.0.1 127.0.0.1 UDP 51 44381 → 54321 Len=9
Frame 1: 48 bytes on wire (384 bits), 48 bytes captured (384 bits) on in
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_0
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
User Datagram Protocol, Src Port: 44381, Dst Port: 54321
Data (6 bytes)
0000 00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00 ..... E
0010 00 22 ee fb 40 00 40 11 4d cd 7f 00 00 01 7f 00 ."..@.@. M.....
0020 00 01 ad 5d d4 31 00 0e fe 21 48 65 6c 6c 6f 0a ...].1.. .!Hello.
wireshark_lo35D6J3.pcapng
Packets: 3 · Displayed: 3 (100.0%)
Profile: Default


Feb 6 15:53
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubunt...
tanmayi-nagabhairava@tanmayi-nagabhairava-ubunt...
tanmayi-nagabhairava@tanmayi-nagabhairava-ubunt...
-u UDP mode
-V rtable Specify alternate routing table
-v Verbose
-W recvlimit Terminate after receiving a number of packets
-w timeout Timeout for connects and final net reads
-X proto Proxy protocol: "4", "5" (SOCKS) or "connect"
-x addr[:port] Specify proxy address and port
-Z DCCP mode
-z Zero-I/O mode [used for scanning]
Port numbers can be individual or ranges: lo-hi [inclusive]
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ nc -u -l 54321
Hello
PES1UG24CS493
CN lab 3
^C
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ nc -u -l 54321
hello
PES1UG24CS493
CN LAB 3
^C
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ nc -u -l 54321
Hello
PES1UG24CS493
CN LAB 3


Feb 6 15:54
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox: ~
tanmayi-nagabhairava@tanmayi-nagabhairava-ubunt...
tanmayi-nagabhairava@tanmayi-nagabhairava-ubunt...
tanmayi-nagabhairava@tanmayi-nagabhairava-ubunt...
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ nc -u 127.0.0.1 54321
Hello
PES1UG24CS493
CN lab 3
^C
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ nc -u 127.0.0.1 54321
hello
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ nc -u 127.0.0.1 54321
hello
PES1UG24CS493
CN LAB 3
^C
tanmayi-nagabhairava@tanmayi-nagabhairava-ubuntuvirtualbox:~$ nc -u 127.0.0.1 54321
Hello
PES1UG24CS493
CN LAB 3

2. UDP Analysis Report

Question: How does UDP differ from TCP and why is it suitable for DNS?

Answer: UDP is a connectionless protocol that lacks the setup delays, retransmissions, and error recovery found in TCP.

Question: Why is UDP suitable for DNS?

Answer: It is ideal for DNS as it provides **fast communication** without the setup delays associated with TCP handshakes.

Question: What are your observations about UDP traffic in Wireshark?

Answer: Based on the captured traffic, UDP appears as immediate data transfers between the source and destination without any connection-setup packets like SYN or ACK. Each message sent from the terminal results in a single UDP segment where the "Length" field in Wireshark specifically matches the size of the text string sent (e.g., 6 bytes for "Hello"). Unlike the TCP capture, there are no extra packets for handshaking or acknowledgment.

Part 3: Analysis of the UDP protocol in DNS

1. Select the first UDP segment in your trace.

- **1. What is the packet number of this segment in the trace file?**

  ANS: The packet number is **1**.

- **2. What type of application-layer payload or protocol message is being carried in this UDP segment?**

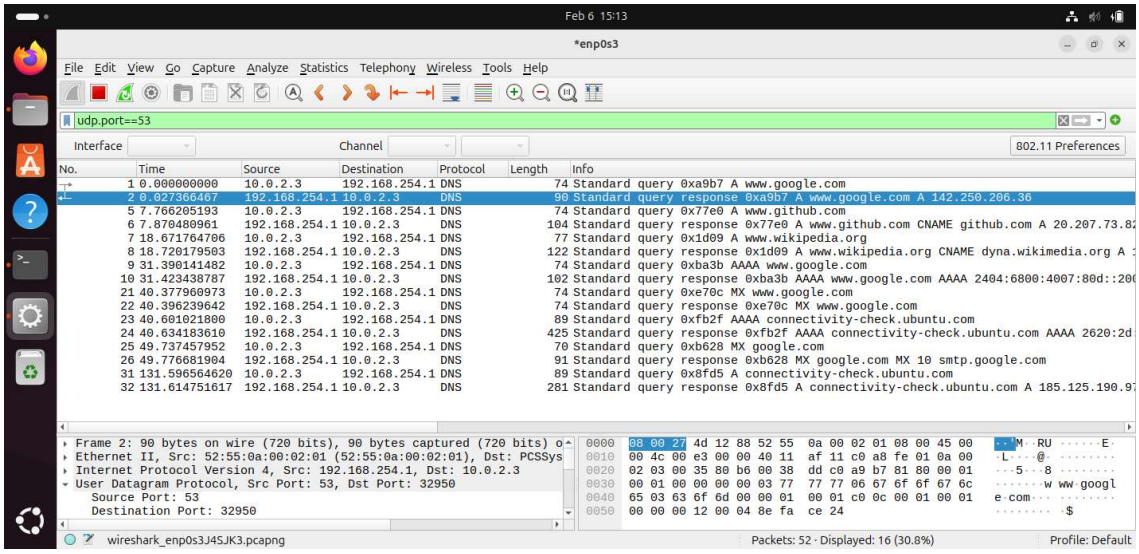
  ANS: It is carrying a **DNS (Domain Name System) query** for www.google.com

- **3. How many fields are there in the UDP header? What are the names of these fields?**

  ANS: There are **4 fields**. The names are **Source Port**, **Destination Port**, **Length**, and **Checksum**.

2. By consulting the displayed information in Wireshark's packet content field for this packet, what is the length (in bytes) of the UDP header field?

ANS: The UDP header is **8 bytes** long. (Each of the four fields is 16 bits, or 2 bytes, totaling 8 bytes).



3. Examine the pair of UDP packets (Query and Reply).

- **A. What is the packet number of these two UDP segments in the trace file?**

  ANS: The packet numbers are **1** (the query) and **2** (the response).

- **B. What is the value in the source port field in this UDP segment (Query)?**

  ANS: The source port is **32950**.

- **C. What is the value in the destination port field in this UDP segment (Query)?**

  ANS: The destination port is **53**.

- **D. What is the value in the source port field in this second UDP segment (Reply)?**

  ANS: The source port is **53**.

- **E. What is the value in the destination port field in this second UDP segment (Reply)?**

  ANS: The destination port is **32950**.

Relationship between the port numbers: The port numbers are **swapped** between the query and the reply. The destination port of the query becomes the source port of the reply, and the source port of the query becomes the destination port of the reply to ensure the data is returned to the correct application.